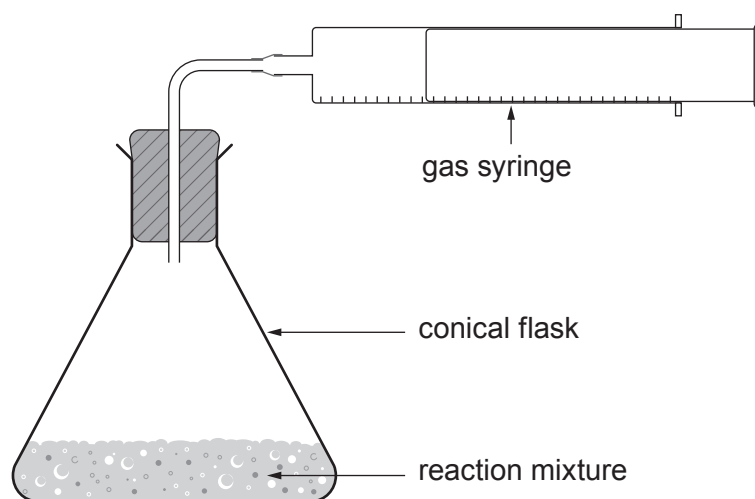


7. A student decided to investigate the rate of reaction of different concentrations of hydrochloric acid with pieces of chalk, using the equipment shown.



Chalk contains calcium carbonate and the reaction produces carbon dioxide gas.

The table shows the student's results.

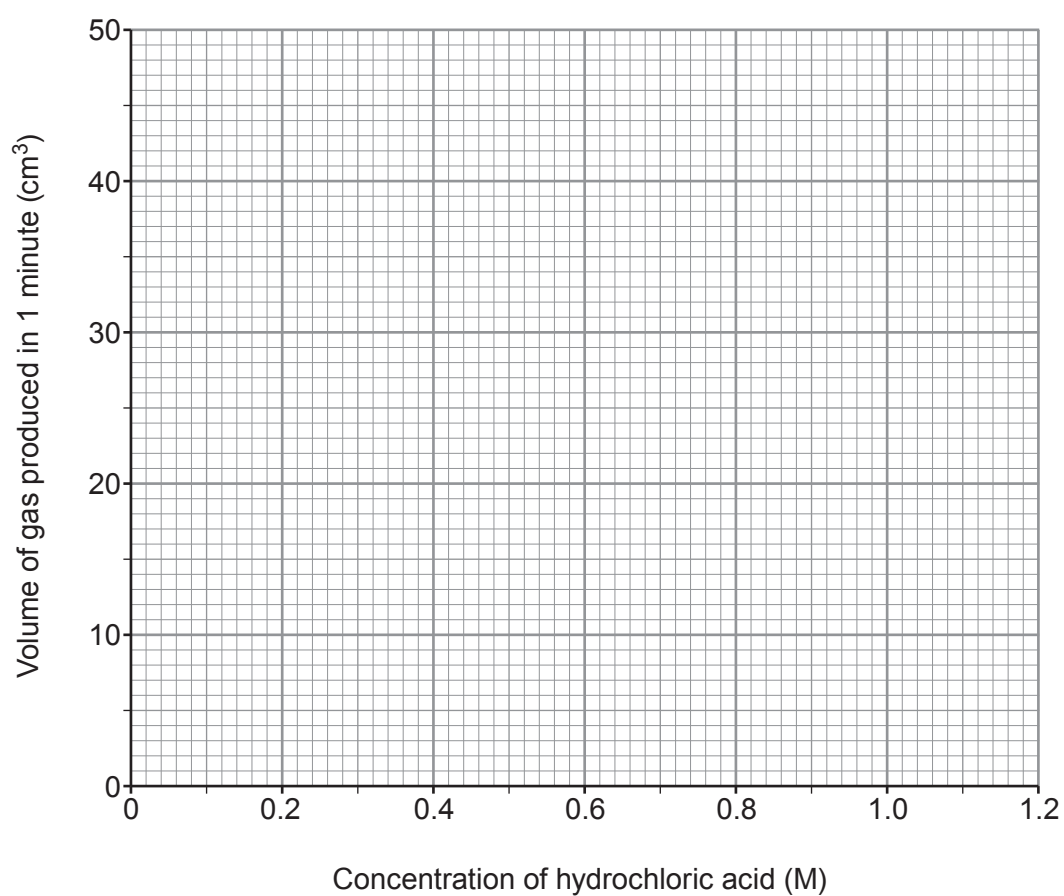
Concentration of hydrochloric acid (M)	Volume of gas produced in 1 minute (cm ³)
0.2	8
0.4	16
0.6	24
0.8	32
1.0	40
1.2	48



(a) Plot the results from the table on the grid. Draw a suitable line.

[3]

Examiner
only



- (b) Describe how the volume of gas produced in 1 minute changes as the concentration of the acid changes. [2]

.....

.....

.....

- (c) Underline the correct word(s) in each bracket to explain the results. [3]

When the acid concentration is higher, there are

(**more** / less / the same number of) particles in the same volume.

Acid particles (**dissolve** / mix / collide) with the chalk more frequently.

The (**chalk** / acid / gas) is produced at a higher rate.

- (d) Other than changing the concentration of the acid or the mass of the chalk added, state **two** ways in which the rate of this reaction could be increased. [2]

.....

.....

.....

